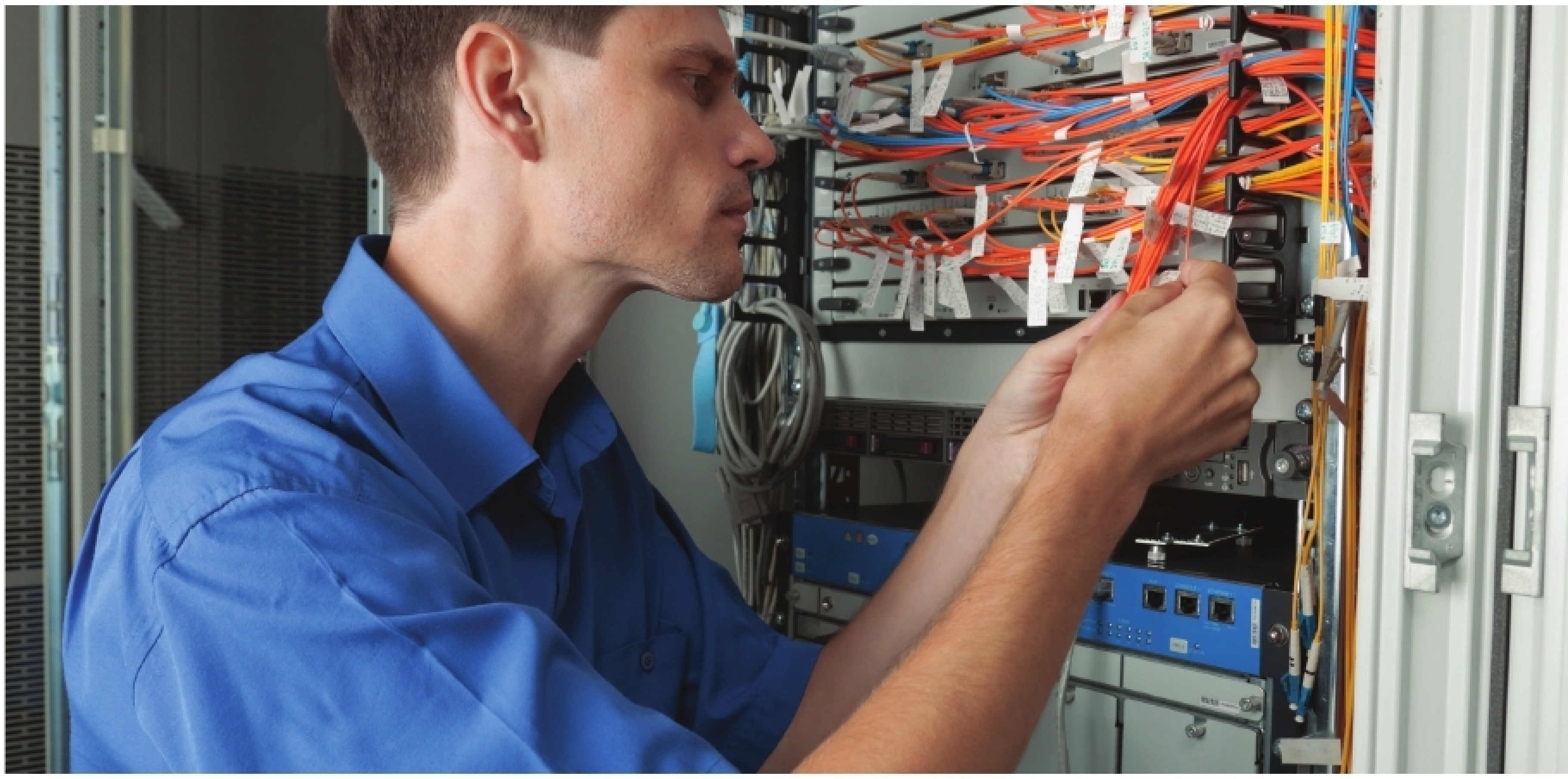
Troubleshooting Network Issues

This article is a continuation of the introductory article on 'Troubleshooting Network Issues' that was carried in the previous issue of *Open Source For You*. Those joining us now are advised to read the previous article to understand this one better.



Linux systems support device names based on hardware attributes. This device-naming approach is implemented on physical systems. It requires the *biosdevname* package to be installed and the *biosdevname=1* kernel parameter to be specified. Hardware vendors enable this, by default, on most systems.

On-board network devices are assigned names similar to *eno1,* where 'en' is the Ethernet prefix, and 'o' stands for 'on-board', followed by a device number. PCI Express hot plug slot devices have names similar to *ens1,* where 's' stands for 'slot'. Other network devices have names based on the physical location of the hardware connector *enp2s0.* Traditional, unpredictable naming, such as *eth0,* is used if the previous methods do not work.

Note: Network devices of virtual guests use traditional network device naming.

The */etc/udev/rules.d/80-net-name-slot.rules* file implements persistent naming. This file is symbolically linked to */dev/null* when traditional device-naming is desired. Systems administrators can also write their own *udev* device-naming rules in */etc/udev/rules.d/70-persistent-net.rules.* These rules override the default network device-naming performed by *systemd.*

Log files play a very important role in troubleshooting network issues. These files provide useful information that helps troubleshoot network device-naming issues. The */var/log/messages* file has device detection information that is read at boot time. It also records the messages that the network manager produces when it tries to activate the device. The example in Figure 1 shows the logs related to the *ens10* interface in */var/log/messages.*

```
grep -n ens10 /var/log/messages
:41:57 abc NetworkManager[803]: <info>  ifcfg-rh: new connection /etc/sysconfig/network-scripts/ifcfg-ens10 (50c78fec
:41:57 abc NetworkManager[803]: <info>  (ens10): new Ethernet device (carrier: OFF, driver: '8139cp', ifindex: 3)
:41:57 abc NetworkManager[803]: <info>  (ens10): device state change: unmanaged -> unavailable (reason 'managed') [10
:41:57 abc kernel: IPv6: ADDRCONF(NETDEV_UP): ens10: link is not ready
:41:57 abc kernel: 8139cp 0000:00:0a.0 ens10: link up, 100Mbps, full-duplex, lpa 0x05E1
:41:57 abc NetworkManager[803]: <info>  (ens10): link connected
:41:57 abc NetworkManager[803]: <info>  (ens10): device state change: unavailable -> disconnected (reason 'none') [20
:41:57 abc NetworkManager[803]: <info>  Auto-activating connection 'ens10'.
:41:57 abc NetworkManager[803]: <info>  (ens10): Activation: starting connection 'ens10' (50c78fec-6902-4b1e-9c79-b73
:41:57 abc NetworkManager[803]: <info>  (ens10): device state change: disconnected -> prepare (reason 'none') [30 40
:41:57 abc NetworkManager[803]: <info>  (ens10): device state change: prepare -> config (reason 'none') [40 50 0]
:41:58 abc NetworkManager[803]: <info>  (ens10): device state change: config -> ip-config (reason 'none') [50 70 0]
:41:58 abc avahi-daemon[708]: Joining mDNS multicast group on interface ens10.IPv4 with address 192.168.123.13.
:41:58 abc avahi-daemon[708]: New relevant interface ens10.IPv4 for mDNS.
:41:58 abc avahi-daemon[708]: Registering new address record for 192.168.123.13 on ens10.IPv4.
:41:58 abc NetworkManager[803]: <info>  (ens10): device state change: ip-config -> ip-check (reason 'none') [70 80 0]
:41:58 abc NetworkManager[803]: <info>  (ens10): device state change: ip-check -> secondaries (reason 'none') [80 90
:41:58 abc NetworkManager[803]: <info>  (ens10): device state change: secondaries -> activated (reason 'none') [90 10
:41:58 abc NetworkManager[803]: <info>  Policy set 'ens10' (ens10) as default for IPv4 routing and DNS.
:41:58 abc NetworkManager[803]: <info>  (ens10): Activation: successful, device activated.
:41:58 abc nm-dispatcher: Dispatching action 'up' for ens10
:41:59 abc avahi-daemon[708]: Registering new address record for fe80::5054:ff:fe60:81a1 on ens10.*.
:42:03 abc network: Bringing up interface ens10:  [  OK  ]
:25:31 abc NetworkManager[804]: <info>  ifcfg-rh: new connection /etc/sysconfig/network-scripts/ifcfg-ens10 (50c78fec
:25:31 abc NetworkManager[804]: <info>  (ens10): new Ethernet device (carrier: OFF, driver: '8139cp', ifindex: 3)
:25:31 abc NetworkManager[804]: <info>  (ens10): device state change: unmanaged -> unavailable (reason 'managed') [10
:25:31 abc kernel: IPv6: ADDRCONF(NETDEV_UP): ens10: link is not ready
:25:31 abc kernel: 8139cp 0000:00:0a.0 ens10: link up, 100Mbps, full-duplex, lpa 0x05E1
:25:31 abc NetworkManager[804]: <info>  (ens10): link connected
:25:31 abc NetworkManager[804]: <info>  (ens10): device state change: unavailable -> disconnected (reason 'none') [20
:25:31 abc NetworkManager[804]: <info>  Auto-activating connection 'ens10'.
:25:31 abc NetworkManager[804]: <info>  (ens10): Activation: starting connection 'ens10' (50c78fec-6902-4b1e-9c79-b73
:25:31 abc NetworkManager[804]: <info>  (ens10): device state change: disconnected -> prepare (reason 'none') [30 40
:25:31 abc NetworkManager[804]: <info>  (ens10): device state change: prepare -> config (reason 'none') [40 50 0]
:25:32 abc NetworkManager[804]: <info>  (ens10): device state change: config -> ip-config (reason 'none') [50 70 0]
```

Figure 1: Logs related to the interface *ens10*